

1) Número de portas abertas identificadas no KillerCoda Ubuntu comando (ip a)

No código ip a foram identificadas 3 interfaces de rede.

```
root@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc fq_codel state UP group default qlen 1000
    link/ether 5a:e3:66:8f:55:c6 brd ff:ff:ff:ff:ff:ff
    inet 172.30.1.2/24 brd 172.30.1.255 scope global dynamic noprefixroute enp1s0
        valid_lft 86307511sec preferred_lft 75518311sec
    inet6 fe80::876b:3d12:58fc:cc3b/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1454 qdisc noqueue state DOWN group default
    link/ether b6:a3:d6:f1:7c:eb brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

2) Serviços em execução em cada porta no KillerCoda Ubuntu comando (ss -tuln)

Netid Process	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*
udp	UNCONN	0	0	172.30.1.2:68	0.0.0.0:*
udp	UNCONN	0	0	172.30.1.2%enp1s0:68	0.0.0.0:*
udp	UNCONN	0	0	[fe80::4bbe:afb6:87ad:b2c]%enp1s0:546	:::*
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*
tcp	LISTEN	0	4096	0.0.0.0:22	0.0.0.0:*
tcp	LISTEN	0	128	0.0.0.0:40200	0.0.0.0:*
tcp	LISTEN	0	511	0.0.0.0:40205	0.0.0.0:*
tcp	LISTEN	0	4096	127.0.0.1:35595	0.0.0.0:*
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*
tcp	LISTEN	0	4096	:::22	:::*
tcp	LISTEN	0	4096	*:40300	*:*
tcp	LISTEN	0	4096	*:40305	*:*

Port 53 (TCP/UDP): DNS (Serviço de nomes local do sistema).

Port 22 (TCP): SSH (Acesso remoto ao terminal).

Port 68 (UDP): DHCP (Configuração automática de IP do servidor).

Port 546 (UDP): DHCPv6 (Configuração automática de IP versão 6).

Port 35595 (TCP): Serviço interno (Restrito ao uso da própria máquina).

3) Versões exatas detetadas pelo Nmap

```

root@ip-10-128-160-168:~# nmap -sV -sC 10.128.175.7
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-15 16:34 UTC
Nmap scan report for ip-10-128-175-7.eu-west-3.compute.internal (10.128.175.7)
Host is up (0.00036s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          FileZilla ftpd
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ Can't get directory listing: TIMEOUT
| ftp-syst:
|_  SYST: UNIX emulated by FileZilla
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
|_ http-methods:
|_  Potentially risky methods: TRACE
|_ http-server-header: Microsoft-IIS/10.0
|_ http-title: IIS Windows Server
135/tcp   open  msrpc        Microsoft Windows RPC
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
|_ ssl-date: 2026-07-15T16:35:38+00:00; 0s from scanner time.
|_ rdp-ntlm-info:
|_   Target_Name: WIN-SCAN
|_   NetBIOS_Domain_Name: WIN-SCAN
|_   NetBIOS_Computer_Name: WIN-SCAN
|_   DNS_Domain_Name: win-scan
|_   DNS_Computer_Name: win-scan
|_   Product_Version: 10.0.17763
|_   System_Time: 2026-07-15T16:35:08+00:00
|_ ssl-cert: Subject: commonName=win-scan
|_ Not valid before: 2026-07-14T16:26:58
|_ Not valid after:  2027-01-13T16:26:58
MAC Address: 06:88:93:B5:9F:3F (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 55.96 seconds

```

Ports	Serviços	Versões Detetadas
21/tcp	ftp	FileZilla ftpd
53/tcp	domain	Simple DNS Plus
80/tcp	http	Microsoft IIS httpd 10.0
135/tcp	msrpc	Microsoft Windows RPC
3389/tcp	ms-wbt-server	Microsoft Terminal Services

4) Output completo do comando ip a e ss -tuln do ambiente local (TryHackMe Further Nmap)

Comando ip a no TryHackMe Further Nmap

```
root@ip-10-128-160-168:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 06:c2:09:e4:4f:f3 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 10.128.160.168/18 metric 100 brd 10.128.191.255 scope global dynamic ens5
        valid_lft 2156sec preferred_lft 2156sec
    inet6 fe80::4c2:9ff:fee4:4ff3/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 0a:5a:b8:0a:0c:fc brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::85a:b8ff:fe0a:cfc/64 scope link
        valid_lft forever preferred_lft forever
4: veth40c46fd@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 7e:19:31:1a:f4:dd brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::7c19:31ff:fe1a:f4dd/64 scope link
        valid_lft forever preferred_lft forever
5: veth0000c82@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether e2:7f:96:5a:3e:88 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::e07f:96ff:fe5a:3e88/64 scope link
        valid_lft forever preferred_lft forever
```

Resultado

- ✓ lo (Loopback): Interface de rede local usada para comunicação interna do sistema, com o endereço IP 127.0.0.1.
- ✓ ens5: A interface de rede principal desta máquina. Com endereço IP dinâmico 10.128.160.168/18 e um endereço IPv6 (fe80::4c2:9ff:fee4:4ff3/64).
- ✓ docker0: Uma interface de rede virtual criada pelo Docker para permitir a comunicação entre contêineres, com o endereço 172.17.0.1/16.
- ✓ Interfaces veth...: São interfaces virtuais (ethernet virtual) conectadas à ponte docker0,

Output completo do comando `ss -tln` no TryHackMe Further Nmap

```
root@ip-10-128-160-168:~# nmap -sV -sC 10.128.175.7
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-15 16:34 UTC
Nmap scan report for ip-10-128-175-7.eu-west-3.compute.internal (10.128.175.7)
Host is up (0.00036s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          FileZilla ftpd
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ Can't get directory listing: TIMEOUT
| ftp-syst:
|_ SYST: UNIX emulated by FileZilla
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
|_ http-methods:
|_ Potentially risky methods: TRACE
|_ http-server-header: Microsoft-IIS/10.0
|_ http-title: IIS Windows Server
135/tcp   open  msrpc        Microsoft Windows RPC
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
|_ ssl-date: 2026-07-15T16:35:38+00:00; 0s from scanner time.
|_ rdp-ntlm-info:
|   Target_Name: WIN-SCAN
|   NetBIOS_Domain_Name: WIN-SCAN
|   NetBIOS_Computer_Name: WIN-SCAN
|   DNS_Domain_Name: win-scan
|   DNS_Computer_Name: win-scan
|   Product_Version: 10.0.17763
|_ System_Time: 2026-07-15T16:35:08+00:00
|_ ssl-cert: Subject: commonName=win-scan
|_ Not valid before: 2026-07-14T16:26:58
|_ Not valid after: 2027-01-13T16:26:58
MAC Address: 06:88:93:B5:9F:3F (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 55.96 seconds
```

Sistema Operacional: Windows Server (versão 10.0.17763, equivalente ao Windows Server 2019).

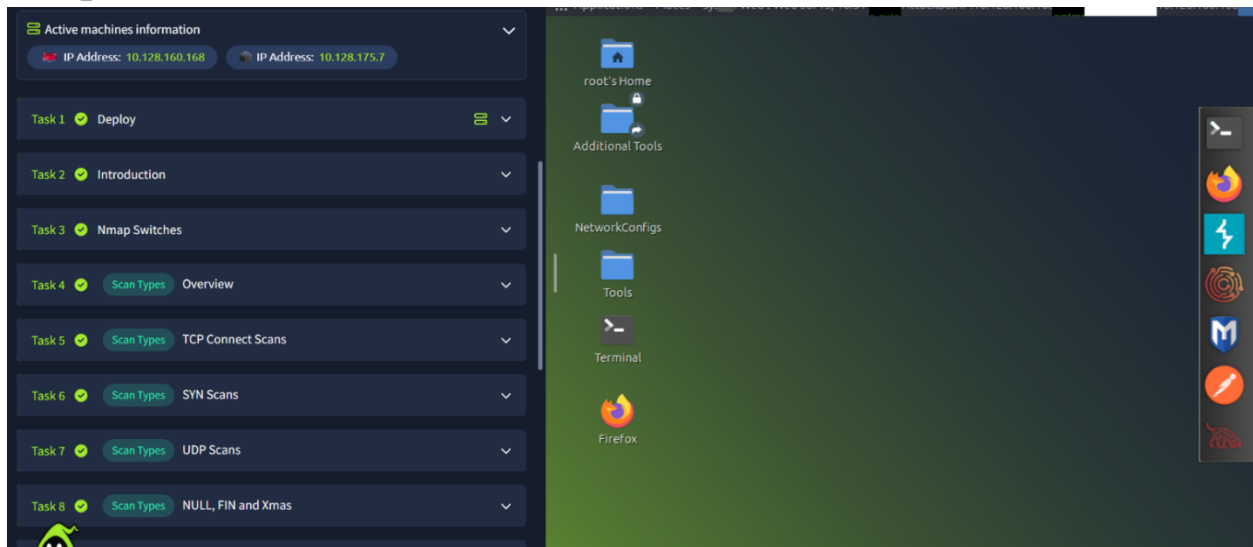
Serviços Ativos:

- ✓ FTP (Porta 21): *FileZilla ftpd*. Permite login anônimo (anon-ftp), o que é uma falha comum de configuração.
- ✓ DNS (Porta 53): *Simple DNS Plus*.
- ✓ HTTP (Porta 80): *Microsoft IIS 10.0*. Servidor web padrão com o método TRACE habilitado (considerado um risco de segurança).
- ✓ RPC (Porta 135): *Microsoft Windows RPC*.
- ✓ RDP (Porta 3389): *Microsoft Terminal Services*. Acesso remoto via Remote Desktop.

Tarefas na sala tryHackMe Further Nmap

```
root@ubuntu:~$ apt install nmap
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'apt autoremove' to remove it.
The following additional packages will be installed:
  libblas3 liblinear4 liblua5.4-0 libssh2-1t64 nmap-common
Suggested packages:
  liblinear-tools liblinear-dev ncat ndiff zenmap
The following NEW packages will be installed:
  libblas3 liblinear4 liblua5.4-0 libssh2-1t64 nmap nmap-common
0 upgraded, 6 newly installed, 0 to remove and 59 not upgraded.
Need to get 6454 kB of archives.
After this operation, 28.0 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Maquinas instaladas



Task 1

Deploy

Your virtual environment has been set up

All machine details can be found at the top of the page.




Attacker machine ⓘ
Status: Connected via Attackbox

Lab machine ⓘ
Status: On [See connection methods](#)

Active machines information

IP Address: 10.128.160.168 IP Address: 10.128.175.7

Task 1

Deploy

Task 2

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Scan Types UDP Scans

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Scan Types NULL, FIN and Xmas

Task 9

Scan Types ICMP Network Scanning

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NSE Scripts Overview

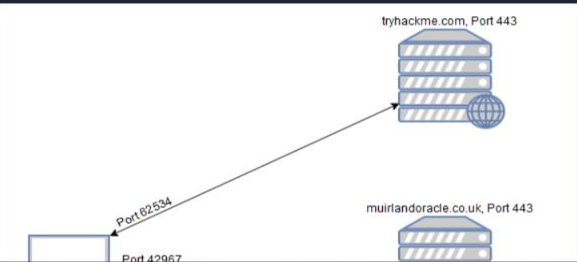
Task links

Task2 — <https://tryhackme.com/room/furtherhmap?taskNo=2&shareId=6a554ac4c3bf3a65a991ef15>

Task 2 Introduction

When it comes to hacking, knowledge is power. The more knowledge you have about a target system or network, the more options you have available. This makes it imperative that proper enumeration is carried out before any exploitation attempts are made.

Say we have been given an IP (or multiple IP addresses) to perform a security audit on. Before we do anything else, we need to get an idea of the "landscape" we are attacking. What this means is that we need to establish which services are running on the targets. For example, perhaps one of them is running a webserver, and another is acting as a Windows Active Directory Domain Controller. The first stage in establishing this "map" of the landscape is something called port scanning. When a computer runs a network service, it opens a networking construct called a "port" to receive the connection. Ports are necessary for making multiple network requests or having multiple services available. For example, when you load several webpages at once in a web browser, the program must have some way of determining which tab is loading which web page. This is done by establishing connections to the remote webserver using different ports on your local machine. Equally, if you want a server to be able to run more than one service (for example, perhaps you want your webserver to run both HTTP and HTTPS versions of the site), then you need some way to direct the traffic to the appropriate service. Once again, ports are the solution to this. Network connections are made between two ports – an open port listening on the server and a randomly selected port on your own computer. For example, when you connect to a web page, your computer may open port 49534 to connect to the server's port 443.



Answer the questions below

What networking constructs are used to direct traffic to the right application on a server?

Ports ✓ Correct Answer

How many of these are available on any network-enabled computer?

65535 ✓ Correct Answer

[Research] How many of these are considered "well-known"? (These are the "standard" numbers mentioned in the task)

1024 ✓ Correct Answer

Task3- <https://tryhackme.com/room/furtherhmap?taskNo=3&shareId=6a554ac4c3bf3a65a991ef15>

Task 3 Nmap Switches

Like most pentesting tools, `nmap` is run from the terminal. There are versions available for both Windows and Linux. For this room we will assume that you are using Linux; however, the switches should be identical. Nmap is installed by default in both Kali Linux and the TryHackMe Attack Box.

Nmap can be accessed by typing `nmap` into the terminal command line, followed by some of the "switches" (command arguments which tell a program to do different things) we will be covering below.

All you'll need for this is the help menu for `nmap` (accessed with `nmap -h`) and/or the `nmap` man page (access with `man nmap`). For each answer, include all parts of the switch unless otherwise specified. This includes the hyphen at the start (`-`).

Answer the questions below

What is the first switch listed in the help menu for a 'Syn Scan' (more on this later!)?

-sS ✓ Correct Answer

Adilson José Furtado

Which switch would you use for a "UDP scan"?

-sU

✓ Correct Answer

If you wanted to detect which operating system the target is running on, which switch would you use?

-O

✓ Correct Answer

Nmap provides a switch to detect the version of the services running on the target. What is this switch?

-sV

✓ Correct Answer

The default output provided by nmap often does not provide enough information for a pentester. How would you increase the verbosity?

-v



✓ Correct Answer

Verbosity level one is good, but verbosity level two is better! How would you set the verbosity level to two?

(Note: it's highly advisable to always use *at least* this option)

-vv

✓ Correct Answer

We should always save the output of our scans -- this means that we only need to run the scan once (reducing network traffic and thus chance of detection), and gives us a reference to use when writing reports for clients.

What switch would you use to save the nmap results in three major formats?

-oA

✓ Correct Answer

What switch would you use to save the nmap results in a "normal" format?

-oN

✓ Correct Answer

Adilson José Furtado

A very useful output format: how would you save results in a "grepable" format?

-oG

✓ Correct Answer

Sometimes the results we're getting just aren't enough. If we don't care about how loud we are, we can enable "aggressive" mode. This is a shorthand switch that activates service detection, operating system detection, a traceroute and common script scanning.

How would you activate this setting?

-A

✓ Correct Answer

Nmap offers five levels of "timing" template. These are essentially used to increase the speed your scan runs at. Be careful though: higher speeds are noisier, and can incur errors!

How would you set the timing template to level 5?

-T5



✓ Correct Answer

We can also choose which port(s) to scan.

How would you tell nmap to only scan port 80?

-p 80

✓ Correct Answer

How would you tell nmap to scan ports 1000-1500?

-p 1000-1500

✓ Correct Answer

A very useful option that should not be ignored:

How would you tell nmap to scan *all* ports?

-p-

✓ Correct Answer

How would you activate a script from the nmap scripting library (lots more on this later!)?

--script

✓ Correct Answer

How would you activate all of the scripts in the "vuln" category?

--script=vuln

✓ Correct Answer



Task 4- <https://tryhackme.com/room/furthernmap?taskNo=4&shareId=6a554ac4c3bf3a65a991ef15>

Task 4 ✓ **Scan Types** Overview

When port scanning with `Nmap`, there are three basic scan types. These are:

- `TCP Connect Scans` (`-sT`)
- `SYN "Half-open" Scans` (`-sS`)
- `UDP Scans` (`-sU`)

Additionally there are several less common port scan types, some of which we will also cover (albeit in less detail). These are:

- `TCP Null Scans` (`-sN`)
- `TCP FIN Scans` (`-sF`)
- `TCP Xmas Scans` (`-sX`)

Most of these (with the exception of `UDP` scans) are used for very similar purposes, however, the way that they work differs between each scan. This means that, whilst one of the first three scans are likely to be your go-to in most situations, it's worth bearing in mind that other scan types exist.

In terms of network scanning, we will also look briefly at `ICMP` (or "ping") scanning.

Answer the questions below

Read the Scan Types Introduction.

✓ Correct Answer

Task 5- <https://tryhackme.com/room/furthernmap?taskNo=5&shareId=6a554ac4c3bf3a65a991ef15>

If, however, the request is sent to an *open* port, the target will respond with a `TCP` packet with the `SYN/ACK` flags set. `Nmap` then marks this port as being *open* (and completes the handshake by sending back a `TCP` packet with `ACK` set).

This is all well and good, however, there is a third possibility.

What if the port is open, but hidden behind a firewall?

Many firewalls are configured to simply **drop** incoming packets. `Nmap` sends a `TCP SYN` request, and receives nothing back. This indicates that the port is being protected by a firewall and thus the port is considered to be *filtered*.

That said, it is very easy to configure a firewall to respond with a `RST TCP` packet. For example, in `IPtables` for Linux, a simple version of the command would be as follows:

```
iptables -I INPUT -p tcp --dport <port> -j REJECT --reject-with tcp-reset
```

This can make it extremely difficult (if not impossible) to get an accurate reading of the target(s).

Answer the questions below

Which RFC defines the appropriate behaviour for the `TCP` protocol?

✓ Correct Answer

?

If a port is closed, which flag should the server send back to indicate this?

✓ Correct Answer

Task 6- <https://tryhackme.com/room/furthernmap?taskNo=6&sharerId=6a554ac4c3bf3a65a991ef15>

Answer the questions below

There are two other names for a SYN scan, what are they?

Half-Open, Stealth

✓ Correct Answer

Can Nmap use a SYN scan without Sudo permissions (Y/N)?

N

✓ Correct Answer

Task 7- <https://tryhackme.com/room/furthernmap?taskNo=7&sharerId=6a554ac4c3bf3a65a991ef15>

Active machines information

IP Address: 10.128.175.7

When a packet is sent to an open `UDP` port, there should be no response. When this happens, Nmap refers to the port as being `open|filtered`. In other words, it suspects that the port is open, but it could be firewalled. If it gets a `UDP` response (which is very unusual), then the port is marked as `open`. More commonly there is no response, in which case the request is sent a second time as a double-check. If there is still no response then the port is marked `open|filtered` and Nmap moves on.

When a packet is sent to a `closed UDP` port, the target should respond with an ICMP (ping) packet containing a message that the port is unreachable. This clearly identifies closed ports, which Nmap marks as such and moves on.

Due to this difficulty in identifying whether a `UDP` port is actually open, `UDP` scans tend to be incredibly slow in comparison to the various TCP scans (in the region of 20 minutes to scan the first 1000 ports, with a good connection). For this reason it's usually good practice to run an Nmap scan with `--top-ports <number>` enabled. For example, scanning with `nmap -sU --top-ports 20 <target>`. Will scan the top 20 most commonly used `UDP` ports, resulting in a much more acceptable scan time.

When scanning `UDP` ports, Nmap usually sends completely empty requests -- just raw `UDP` packets. That said, for ports which are usually occupied by well-known services, it will instead send a protocol-specific payload which is more likely to elicit a response from which a more accurate result can be drawn.

Answer the questions below

If a UDP port doesn't respond to an Nmap scan, what will it be marked as?

open|filtered

✓ Correct Answer

When a UDP port is closed, by convention the target should send back a "port unreachable" message. Which protocol would it use to do so?

ICMP

✓ Correct Answer

Task 8- <https://tryhackme.com/room/furthernmap?taskNo=7&sharerId=6a554ac4c3bf3a65a991ef15>

blocking new connection initiation requests). By sending requests which do not contain the SYN flag, we effectively bypass this kind of firewall. Whilst this is good in theory, most modern IDS solutions are savvy to these scan types, so don't rely on them to be 100% effective when dealing with modern systems.

Answer the questions below

Which of the three shown scan types uses the URG flag?

xmas

✓ Correct Answer

Why are NULL, FIN and Xmas scans generally used?

Firewall Evasion

✓ Correct Answer

Which common OS may respond to a NULL, FIN or Xmas scan with a RST for every port?

Microsoft Windows

✓ Correct Answer

Task 9- <https://tryhackme.com/room/furthernmap?taskNo=9&sharerId=6a554ac4c3bf3a65a991ef15>

Active machines information IP Address: 10.128.160.168 IP Address: 10.128.175.7

the specified network. When it receives a response, it marks the IP address that responded as being alive. For reasons we'll see in a later task, this is not always accurate; however, it can provide something of a baseline and thus is worth covering.

To perform a ping sweep, we use the `-sn` switch in conjunction with IP ranges which can be specified with either a hyphen (-) or CIDR notation. i.e. we could scan the `192.168.0.x` network using:

- `nmap -sn 192.168.0.1-254`

or

- `nmap -sn 192.168.0.0/24`

The `-sn` switch tells Nmap not to scan any ports -- forcing it to rely primarily on ICMP echo packets (or ARP requests on a local network, if run with sudo or directly as the root user) to identify targets. In addition to the ICMP echo requests, the `-sn` switch will also cause `nmap` to send a `TCP SYN` packet to port 443 of the target, as well as a `TCP ACK` (or `TCP SYN` if not run as root) packet to port 80 of the target.

Answer the questions below

How would you perform a ping sweep on the 172.16.x.x network (Netmask: 255.255.0.0) using Nmap? (CIDR notation)

`nmap -sn 172.16.0.0/16`

Correct Answer

Task 10- <https://tryhackme.com/room/furtherhmap?taskNo=10&sharerId=6a554ac4c3bf3a65a991ef15>

Active machines information

IP Address: 10.128.160.168 IP Address: 10.128.175.7

- **safe** :- Won't affect the target
- **Intrusive** :- Not safe: likely to affect the target
- **vuln** :- Scan for vulnerabilities
- **exploit** :- Attempt to exploit a vulnerability
- **auth** :- Attempt to bypass authentication for running services (e.g. Log into an FTP server anonymously)
- **brute** :- Attempt to bruteforce credentials for running services
- **discovery** :- Attempt to query running services for further information about the network (e.g. query an SNMP server).

A more exhaustive list can be found [here](#).

In the next task we'll look at how to interact with the NSE and make use of the scripts in these categories.

Answer the questions below

What language are NSE scripts written in?

Lua

✓ Correct Answer

Which category of scripts would be a very bad idea to run in a production environment?

Intrusive

✓ Correct Answer

Task 11- <https://tryhackme.com/room/furtherhmap?taskNo=11&sharerId=6a554ac4c3bf3a65a991ef15>

Task 11

NSE Scripts

Working with the NSE

In Task 3 we looked very briefly at the `--script` switch for activating NSE scripts from the `vuln` category using `--script=vuln`. It should come as no surprise that the other categories work in exactly the same way. If the command `--script=safe` is run, then any applicable safe scripts will be run against the target (Note: only scripts which target an active service will be activated).

To run a specific script, we would use `--script=<script-name>`, e.g. `--script=http-fileupload-exploiter`.

Multiple scripts can be run simultaneously in this fashion by separating them by a comma. For example: `--script=smb-enum-users,smb-enum-shares`.

Some scripts require arguments (for example, credentials, if they're exploiting an authenticated vulnerability). These can be given with the `--script-args` Nmap switch. An example of this would be with the `http-put` script (used to upload files using the PUT method). This takes two arguments: the URL to upload the file to, and the file's location on disk. For example:

```
nmap -p 80 --script http-put --script-args http-put.url='dav/shell.php',http-put.file='./shell.php'
```

Note that the arguments are separated by commas, and connected to the corresponding script with periods (i.e. `<script-name>.<argument>`).

A full list of scripts and their corresponding arguments (along with example use cases) can be found [here](#).

Nmap scripts come with built-in help menus, which can be accessed using `nmap --script-help <script-name>`. This tends not to be as extensive as in the link given above, however, it can still be useful when working locally.

Answer the questions below

What optional argument can the `ftp-anon.nse` script take?

maxlist

✓ Correct Answer

Task 12.

<https://tryhackme.com/room/furthernmap?taskNo=12&sharerId=6a554ac4c3bf3a65a991ef15>

Answer the questions below

Search for "smb" scripts in the `/usr/share/nmap/scripts/` directory using either of the demonstrated methods. What is the filename of the script which determines the underlying OS of the SMB server?

smb-os-discovery.nse

✓ Correct Answer

Read through this script. What does it depend on?

smb-brute

✓ Correct Answer



Task 13

<https://tryhackme.com/room/furthernmap?taskNo=13&sharerId=6a554ac4c3bf3a65a991ef15>

Answer the questions below

Which simple (and frequently relied upon) protocol is often blocked, requiring the use of the `-Pn` switch?

ICMP

✓ Correct Answer

[Research] Which Nmap switch allows you to append an arbitrary length of random data to the end of packets?

--data-length

✓ Correct Answer